Nion Rahaman Akash

B.Sc. in Mathematics | Postgraduate Diploma in Data Science

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RESEARCH INTERESTS

- Machine Learning
- Health Care
- Trustworthy and Reliable AI
- Applied and Computational Mathematics

EDUCATION

Postgraduate Diploma (PGD) in Data Science — 2024

United International University (UIU), Bangladesh • CGPA 3.44/4.00

- **Key areas:** Python, Machine Learning, Deep Learning, SQL, Data Analysis, Data Visualization

Bachelor of Science (B.Sc.) in Mathematics — 2016 – 2022

Department of Mathematics, University of Rajshahi, Bangladesh • CGPA 2.96/4.00

Major Theory Courses

- Theory of Statistics
- Statistical Methods and Demography
- Advanced Calculus
- Vector and Tensor Analysis
- Ordinary and Partial Differential Equations
- Computer Programming (Fortran 90)
- Theory of Matrices
- Operations Research
- Real Analysis
- Linear Algebra
- Numerical Analysis
- Number Theory
- Discrete Mathematics and Programming with C

MAJOR PROJECTS

Diabetes Prediction using Machine Learning — View on GitHub

- Built predictive models for diabetes risk using Logistic Regression, Random Forest, and a Keras/TensorFlow neural network
- Performed data preprocessing, feature engineering, model training, and evaluation
- Applied Python-based data science libraries for analysis and experimentation
- Achieved improved performance with ensemble and neural network approaches, reaching 75% accuracy

Other Machine Learning Projects — All code on GitHub

- **Car & Laptop Price Prediction** — regression pipelines (Linear, Ridge, Random Forest, Gradient Boosting) with full data cleaning and cross-validated model comparison
- **Breast Cancer Classification** — Support Vector Machines (SVM) for tumor classification
- **Digit Recognition** — Random Forest and K-Nearest Neighbors (KNN) on handwritten digit data
- **Spam Detection & Human Sentiment Analysis** — NLP (Bag-of-Words, Naive Bayes) for text classification
- **Customer Segmentation** — K-Means Clustering and RFM Analysis for customer behavior grouping

CERTIFICATIONS

- R for Applied Statistics and Data Science — ISRT, University of Dhaka (36-hour intensive)
- Python for Data Science, AI & Development — Coursera

TECHNICAL SKILLS

- **Programming:** Python, R, C, Fortran, MATLAB
- **Machine Learning:** Regression, Classification, Clustering, Model Evaluation
- **Data Science Libraries:** NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn
- **Data Preparation:** Feature scaling, encoding techniques, outlier detection (IQR, Z-score)

EXTRACURRICULAR & LEADERSHIP ACTIVITIES

Volunteer, Transparency International Bangladesh (TIB) **2021 – 2023**

- Conducted primary data collection across diverse rural and urban households to support evidence-based social accountability research
- Promoted institutional transparency and ethical governance in public sectors (Health, Education) through targeted anti-corruption initiatives
- Facilitated information dissemination campaigns, translating complex governance protocols into accessible insights for non-technical citizens

Active Member, University of Rajshahi English Speaking Club

- Facilitated structured communication workshops, enhancing the public speaking and technical articulation skills of diverse peer groups
- Led collaborative learning sessions, fostering an environment of constructive feedback and continuous improvement in academic fluency

STANDARDIZED TEST

International English Language Testing System (IELTS) **28 Jan 2026**

- Overall **6.5** (Listening 7.5 | Reading 6.5 | Writing 6.0 | Speaking 6.0)

REFERENCES

Dr. MD. Sahadat Hossain
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